IMPLEMENTATION AND PERFORMANCE COMPARISON OF CMOS FA AND APPROXIMATE FULL ADDER

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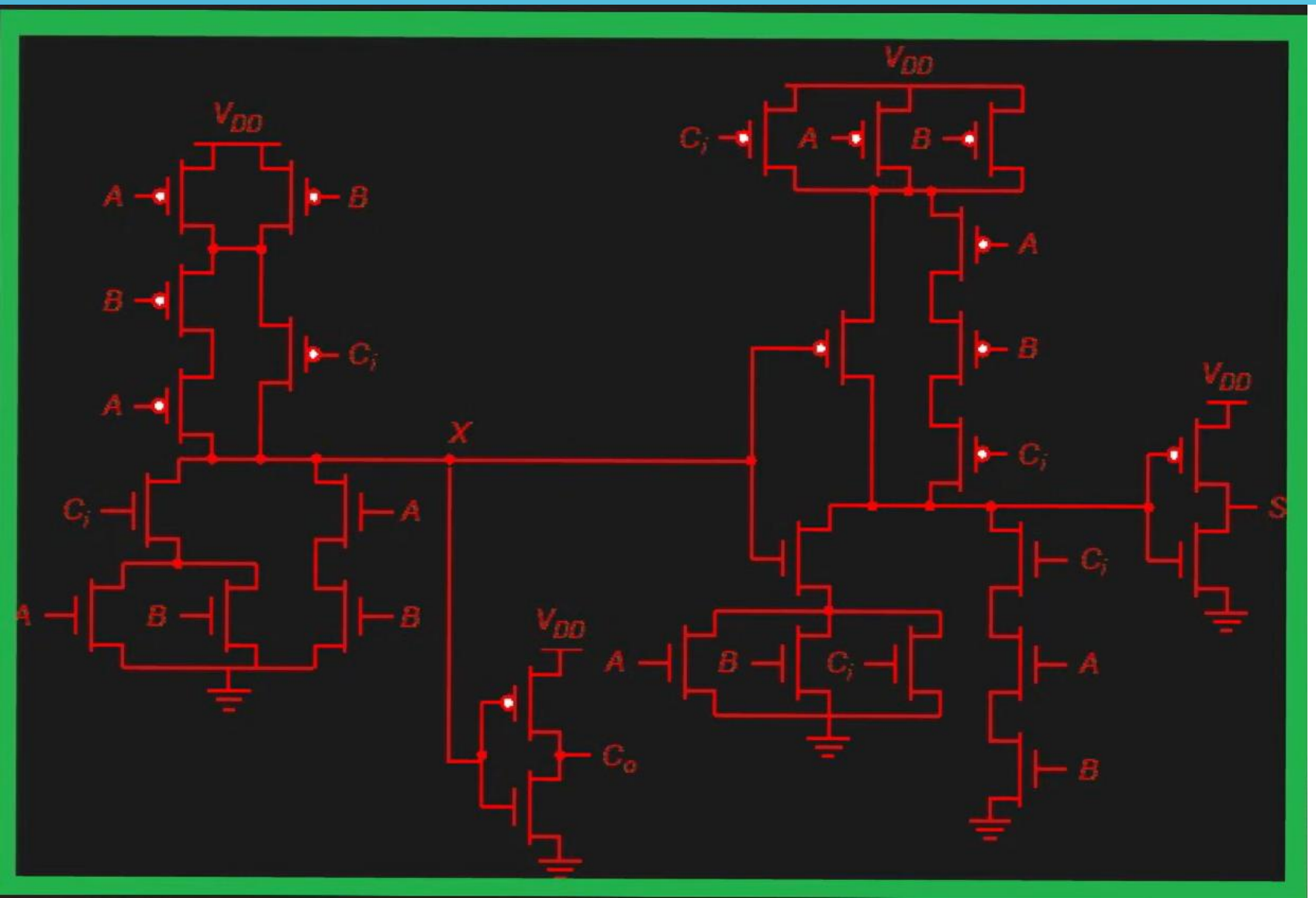
DIGITAL ASSIGNMENT-1
IMPLEMENTATION AND VERIFICATION OF STATIC CMOS FA AND PERFORMANCE ANALYSIS (DELAY AND POWER)

1)CIRCUIT DIAGRAM AND TRUTHTABLE
2)CIRCUIT SCHEMATIC
3)OUTPUT PLOT
4)DELAY AND POWER VALUES

1)CIRCUIT DESIGN AND TRUTH TABLE:

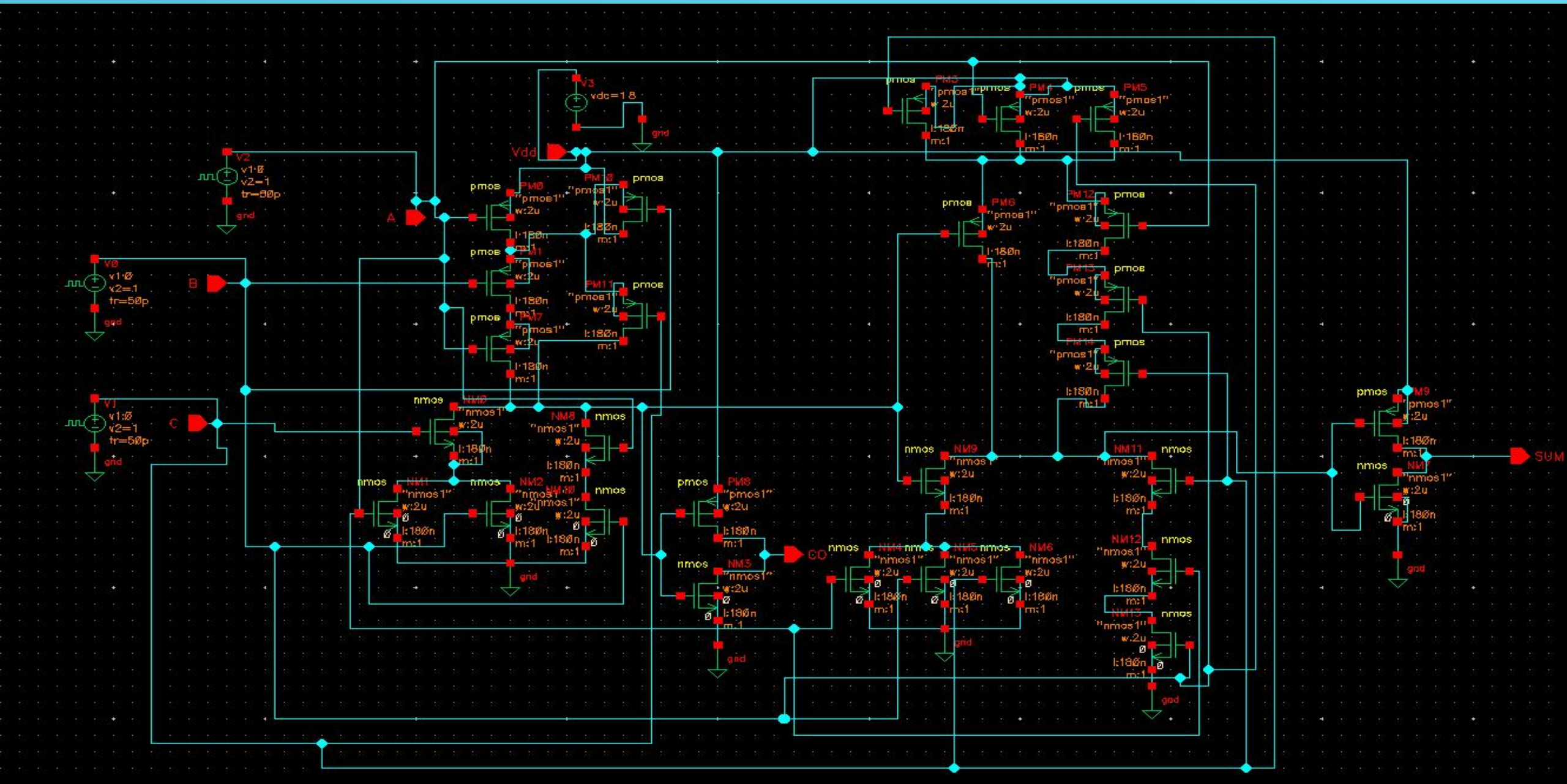
$$S = A \oplus B \oplus C_i$$

$$C_0 = AB + BC_i + AC_i$$



A	B	Cin	Sum (S)	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

2)CIRCUIT SCHEMATIC:



3)OUTPUT PLOT:



4) DELAY AND POWER VALUES:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
1	Expression	average(VT("/SUM"))															
2	Value	8.83E-01															
3	Expression	average(VT("/CO"))															
4	Value	8.83E-01															
5	Expression	delay(?wf1	?value1 0.	?edge1 "ri	?nth1 1	?td1 0.0	?tol1 nil	?wf2 VT("	?value2 0.	?edge2 "ri	?nth2 1	?tol2 nil	?td2 nil	?stop nil	?multiple nil)			
6	Value	1.37E-10															
7	Expression	delay(?wf1	?value1 0.	?edge1 "ri	?nth1 1	?td1 0.0	?tol1 nil	?wf2 VT("	?value2 0.	?edge2 "ri	?nth2 1	?tol2 nil	?td2 nil	?stop nil	?multiple nil)			
8	Value	1.04E-08															
9																	

CONCLUSIONS MADE FROM DA1:

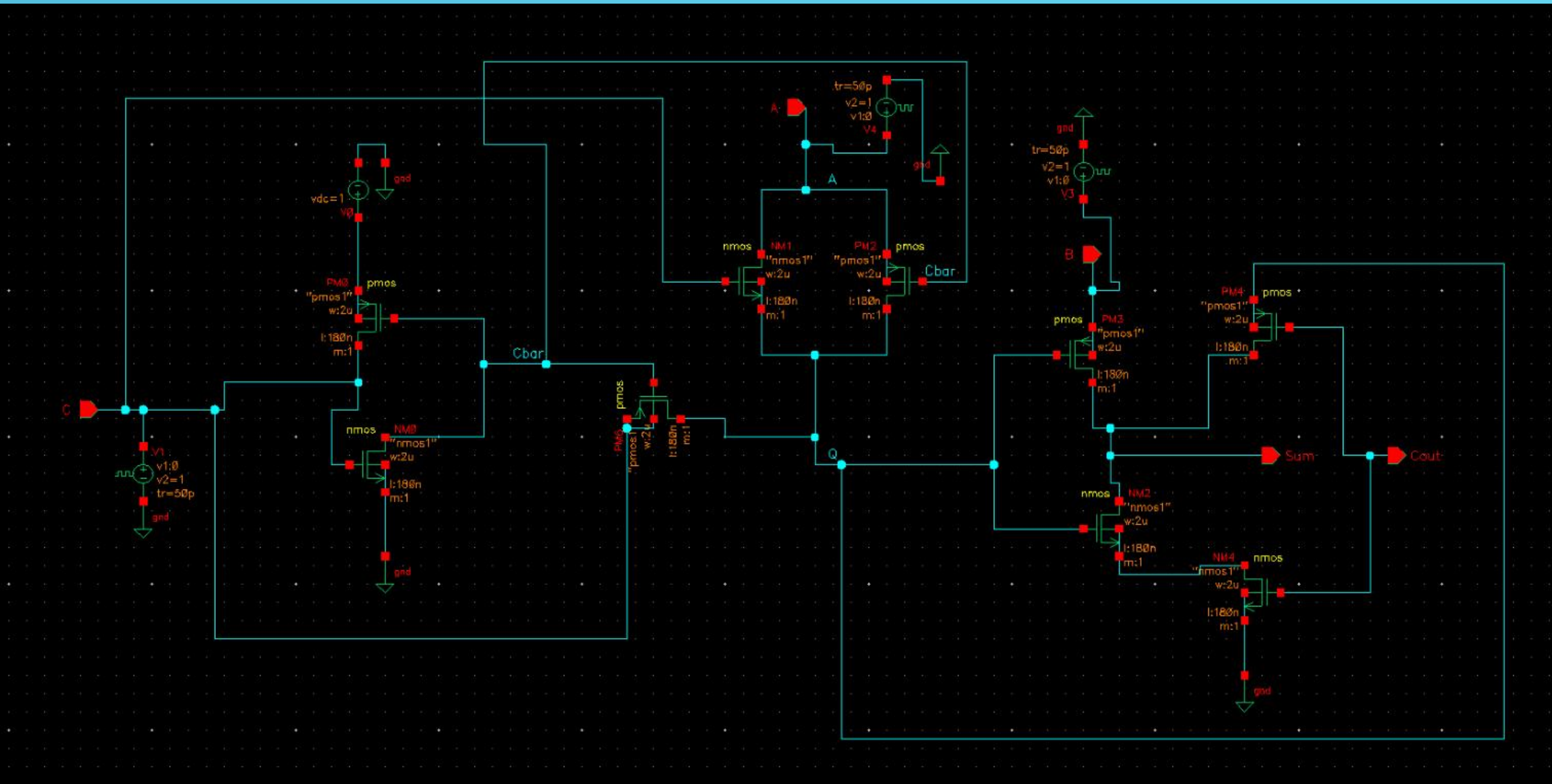
1)Optimized XOR Gates
2)Compact Carry Logic
3)Elimination of Redundant Inverters
4)Mixed Logic Styles
5)No Logic Duplication
6)Ensures correct logical operation with validated output waveforms,
7)Reduces propagation delay through optimized transistor paths
8)Minimizes power consumption using CMOS technology
9)Demonstrates suitability for scalable, low-power digital VLSI systems.

DIGITAL ASSIGNMENT-2
DESIGN AND IMPLEMENTATION OF APPROXIMATE FULL ADDER

FIGURE:(A) LCAFA3,
(B) LCAFA4

1)CIRCUIT DESIGN
2)TRUTH TABLE
3)SIMULATION WAVEFORMS

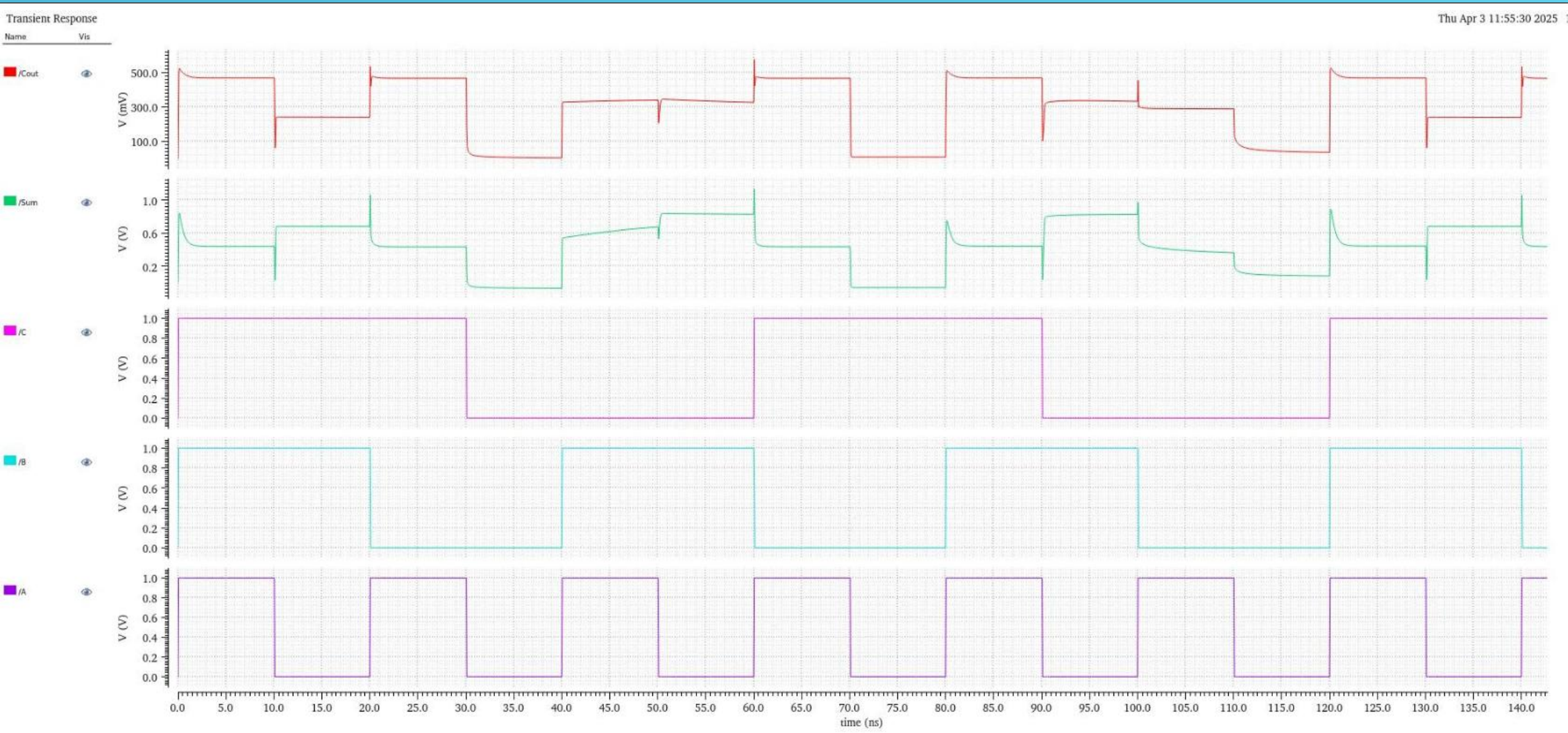
1)(I)CIRCUIT DESIGN(LCAFA3)



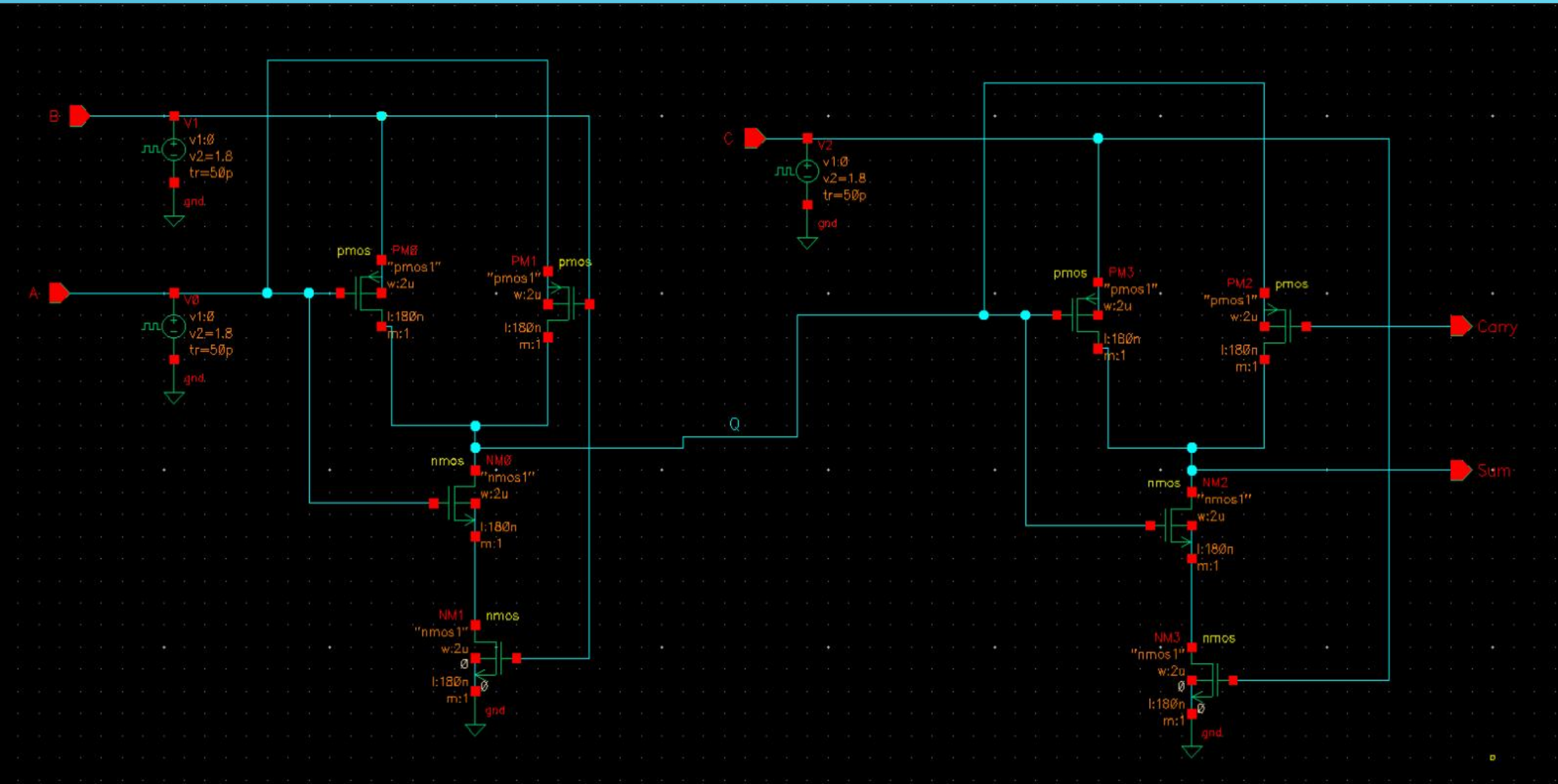
2)(I)TRUTH TABLE:

Input			LCAFA3	
A	B	C	Sum	Cout
0	0	0	0	0
0	0	1	1	
0	1	0	1	0
0	1	1		1
1	0	0	1	0
1	0	1		1
1	1	0	0	
1	1	1	1	1

3)(I)SIMULATION WAVEFORMS:



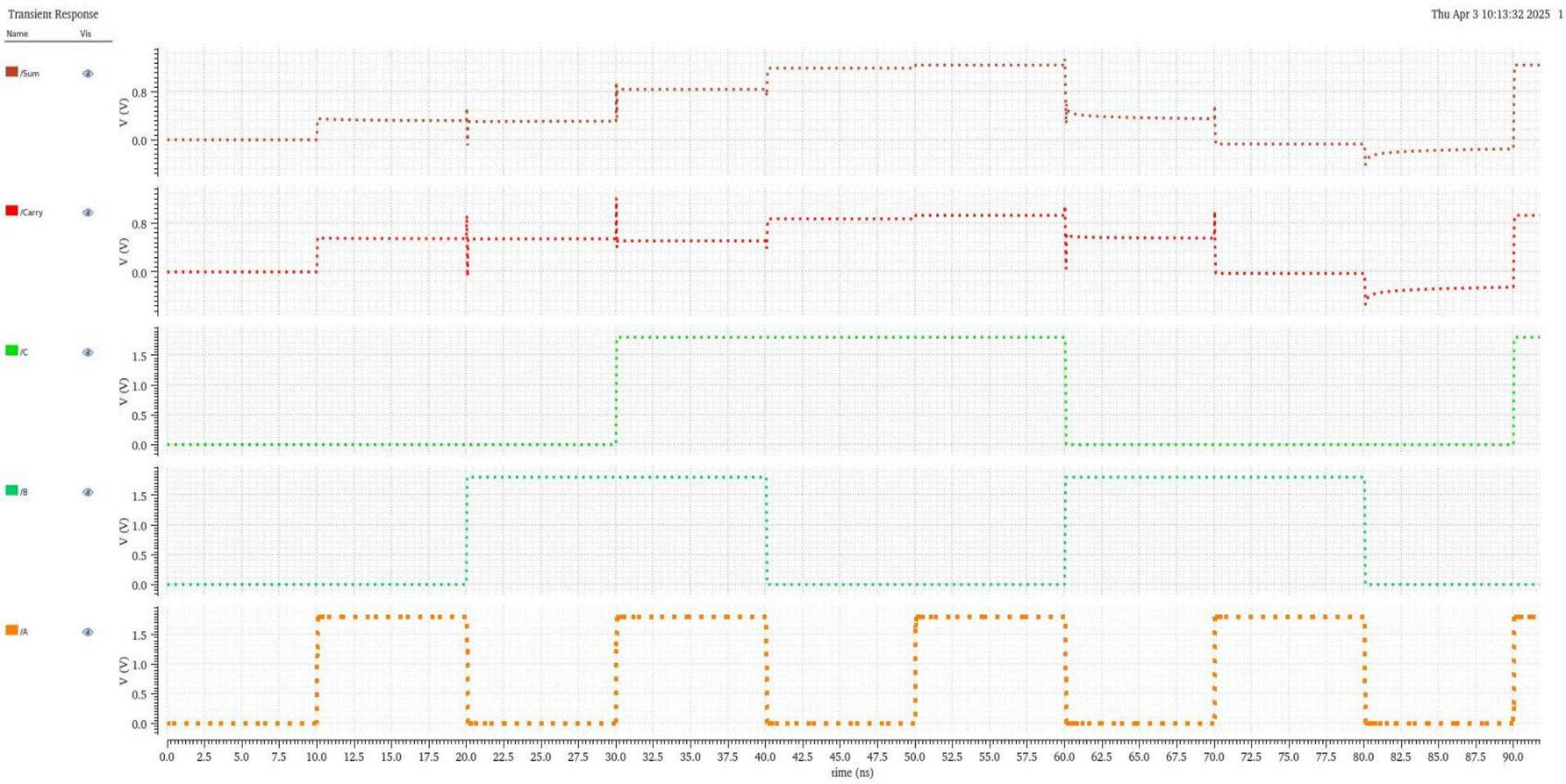
1)(II)CIRCUIT DESIGN(LCAFA4)



2)(II)TRUTH TABLE:

Input			LCAFA4	
A	B	C	Sum	Cout
0	0	0	0	0
0	0	1	1	
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	
1	1	1	1	1

3)(II)SIMULATION WAVEFORMS:



COMPARISON OF STATIC FA AND APPROXIMATE FA:

1)**Complexity & transistor count**: Approximate full adders use fewer transistors than static full adders, reducing area and power consumption.

2) **Accuracy**: Unlike static full adders which are fully accurate, approximate adders may produce errors in certain input conditions to gain efficiency.

3) **Power & delay**: Approximate adders offer lower power consumption and faster operation due to simpler logic paths and fewer switching activities.

4) **Applications**: Approximate adders are ideal for error-tolerant applications like image processing or machine learning, while static adders are used in accuracy-critical systems.

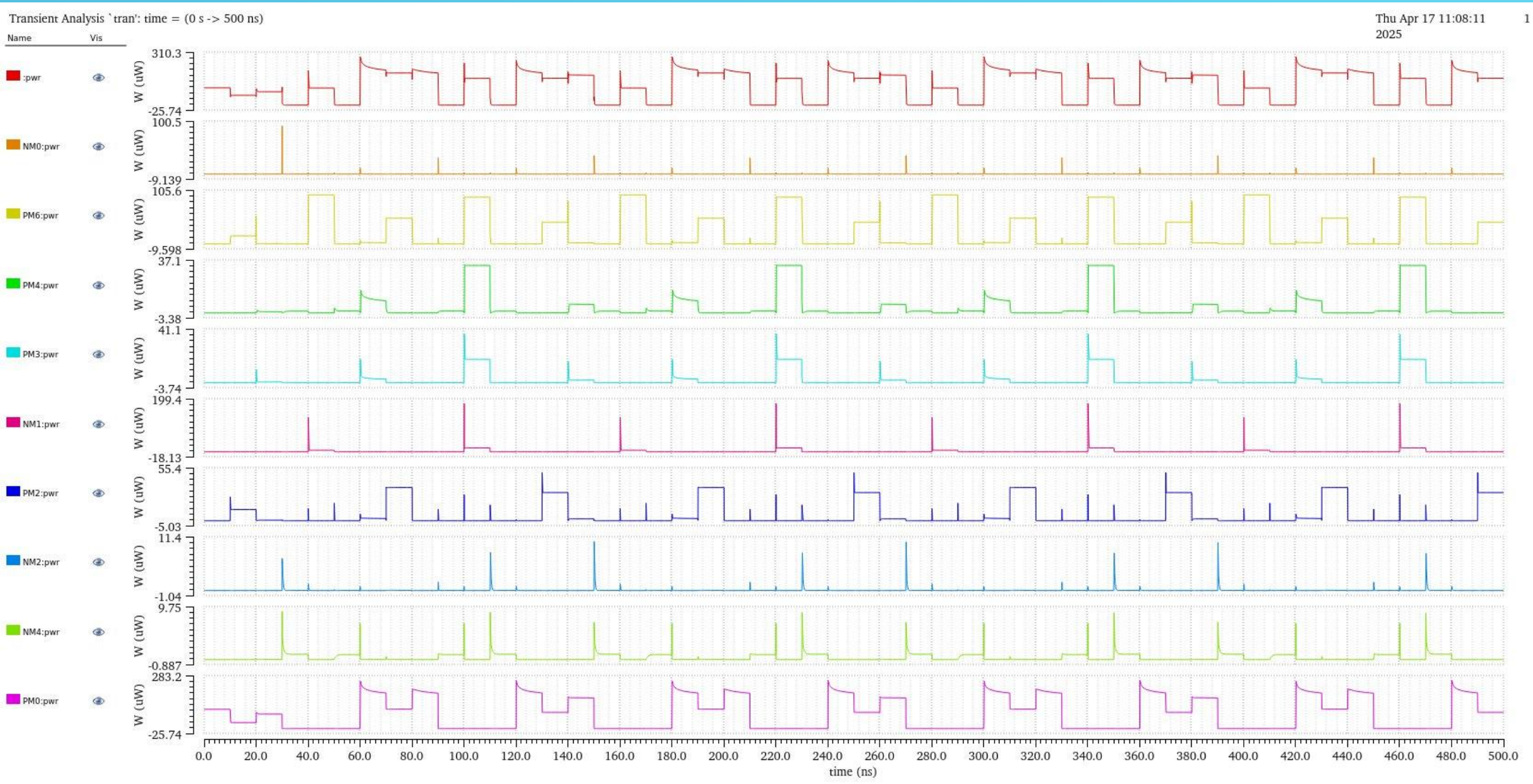
5) **Design style**: Static full adders use complementary pull-up and pull-down networks (cmos logic), whereas the approximate designs often simplify or omit some paths to save resources.

DIGITAL ASSIGNMENT-3

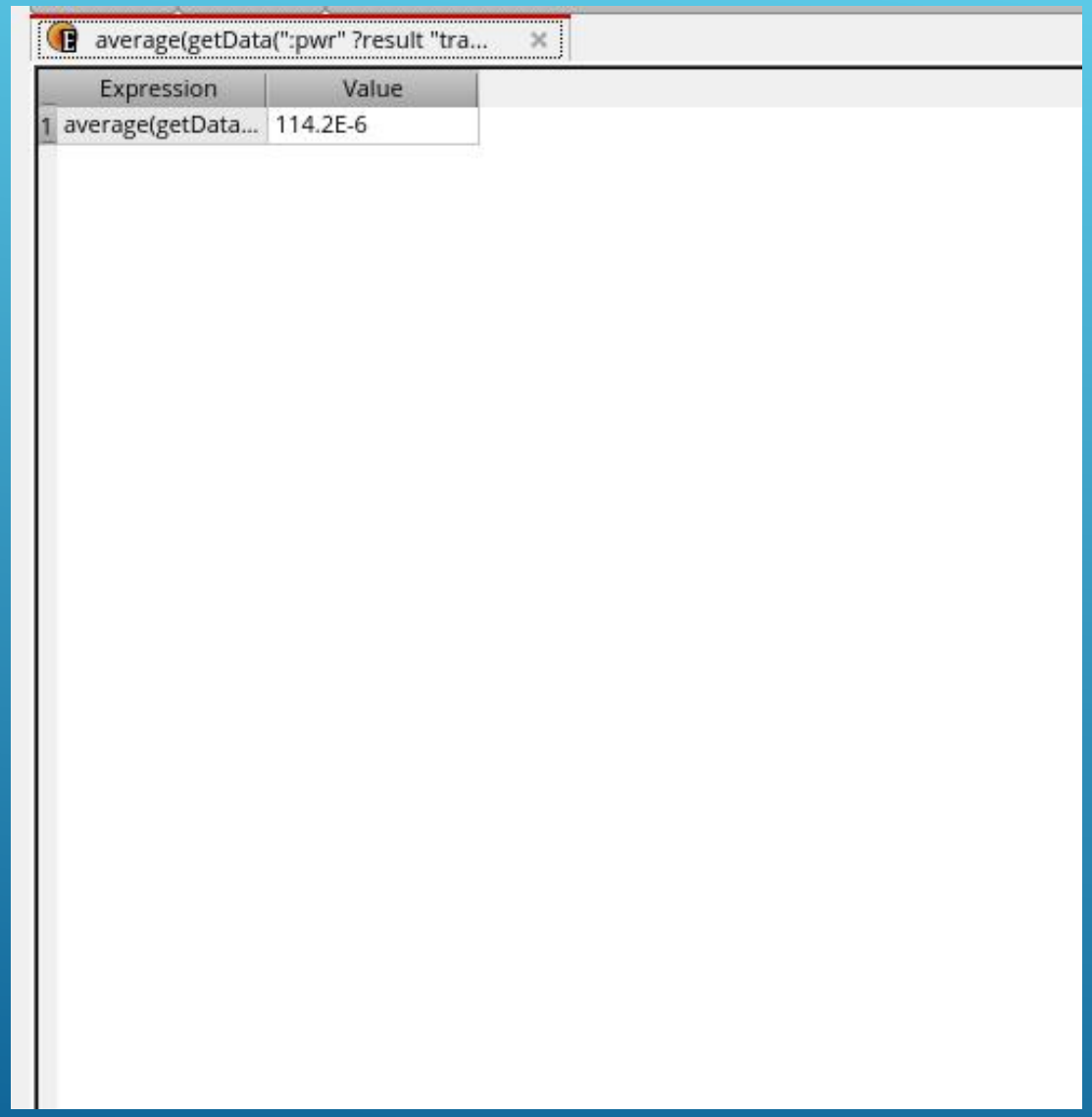
PERFORMANCE ANALYSIS OF APPROXIMATE FA'S (DELAY AND POWER)

1)DELAY AND POWER ANALYSIS OF LCAFA3,LCAFA4

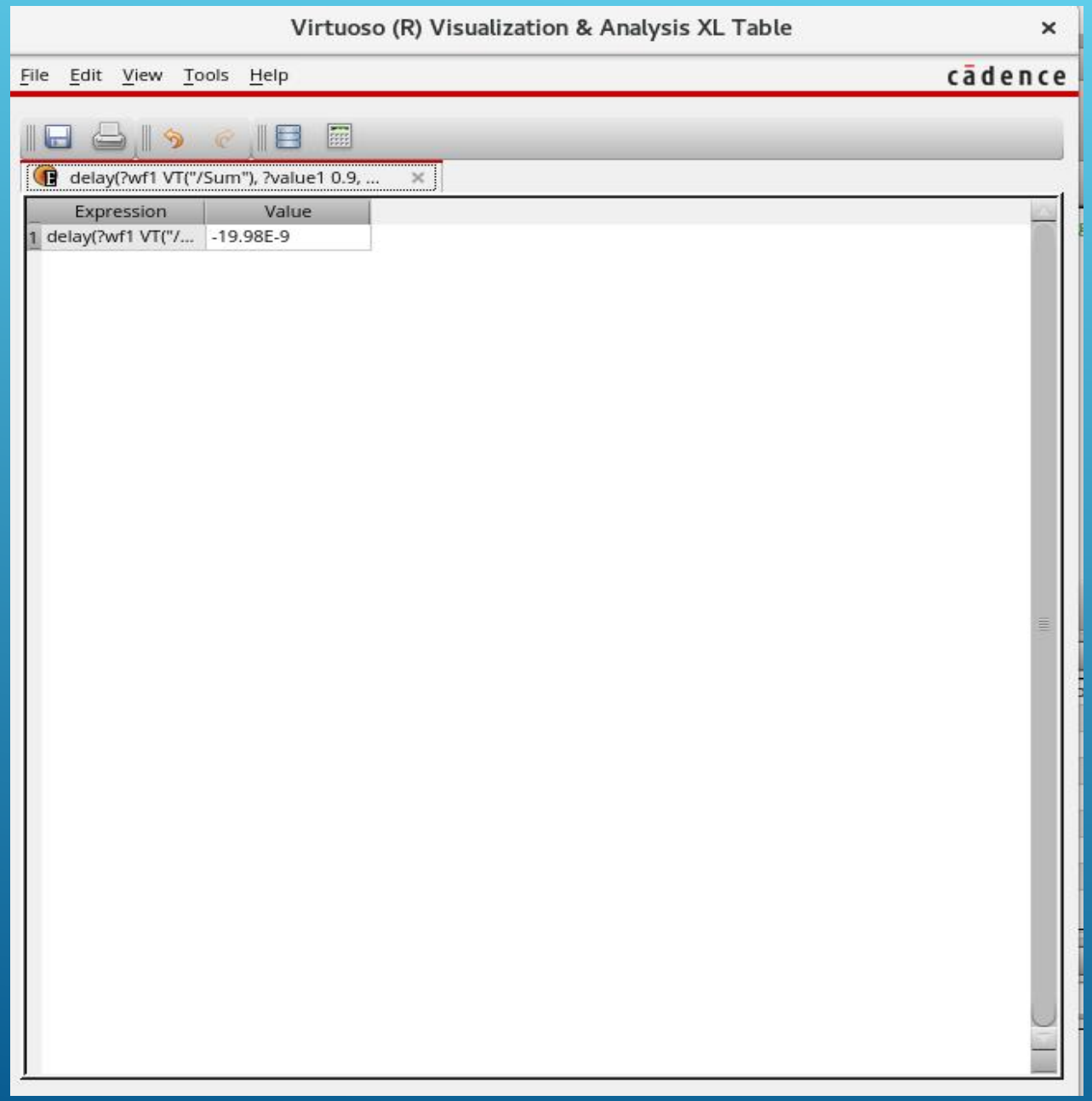
1)(I)POWER GRAPH:

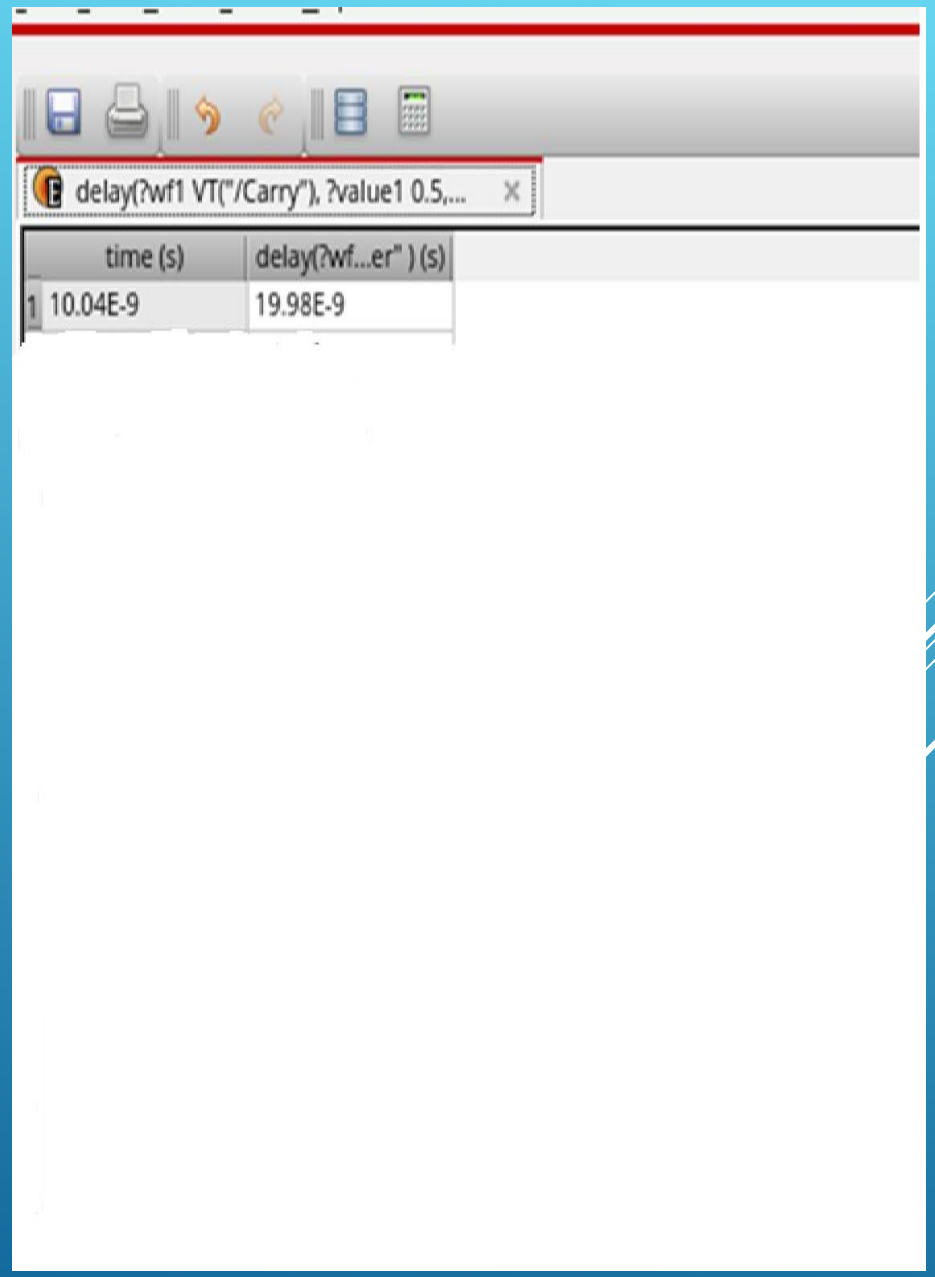


2)(I)POWER VALUE:

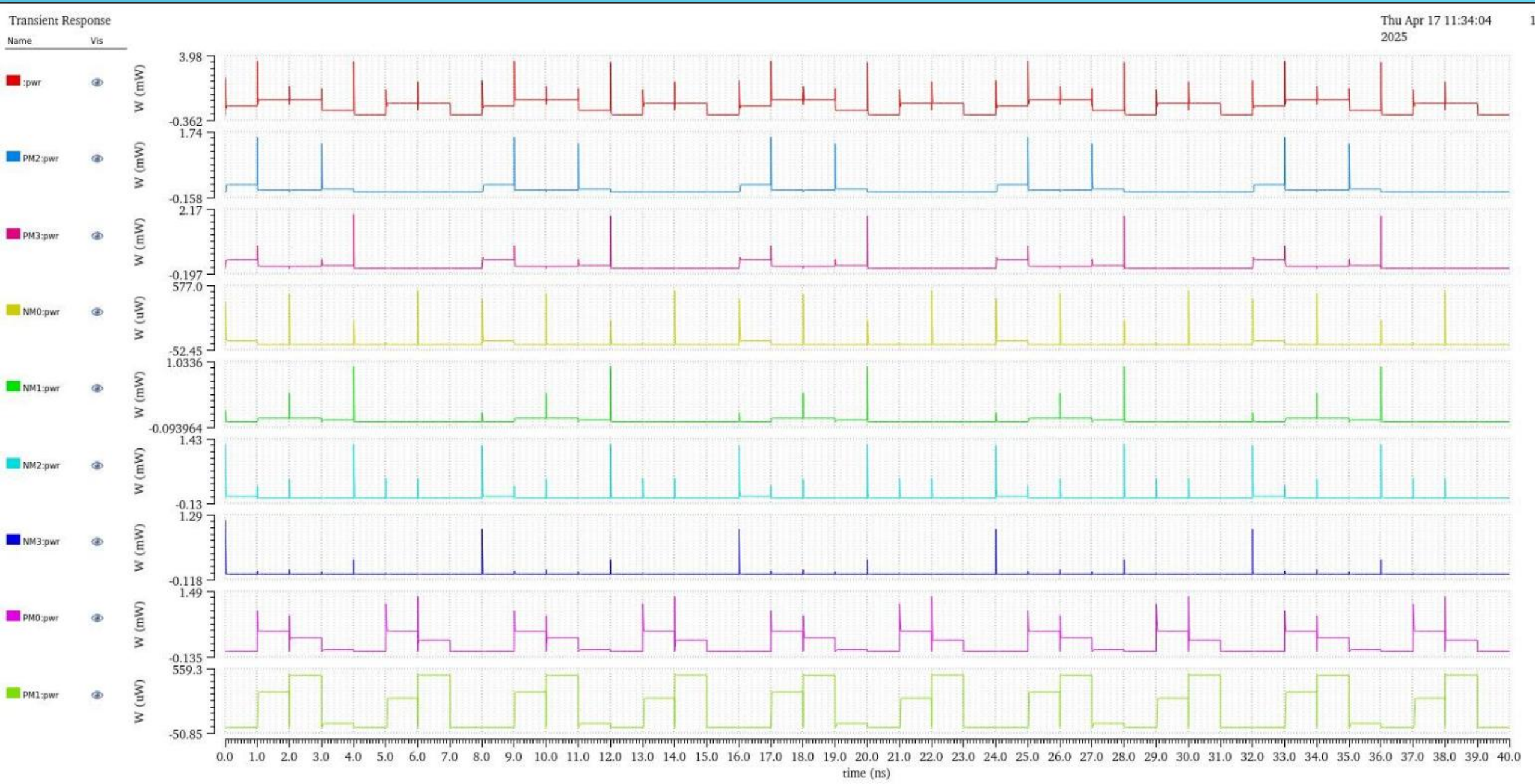


3(I)DELAY VALUE:

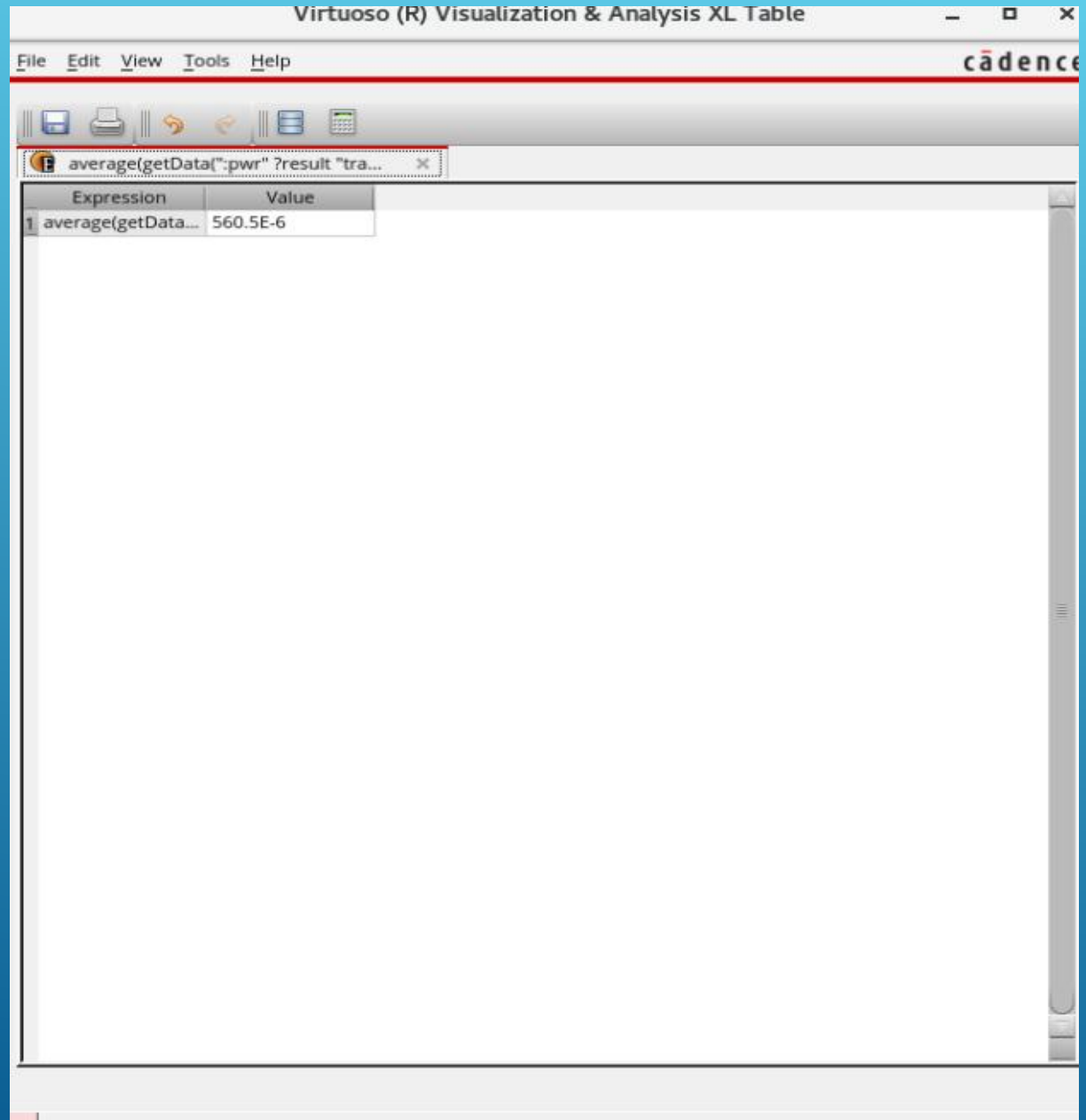




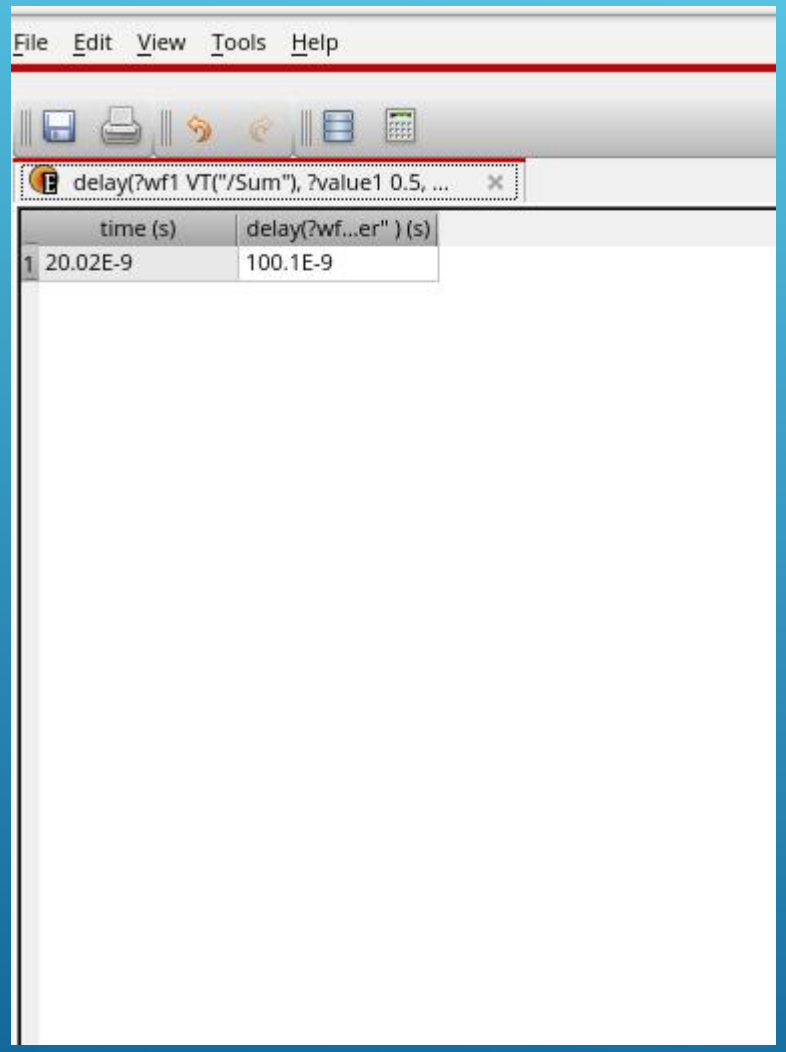
1)(II)POWER GRAPH:

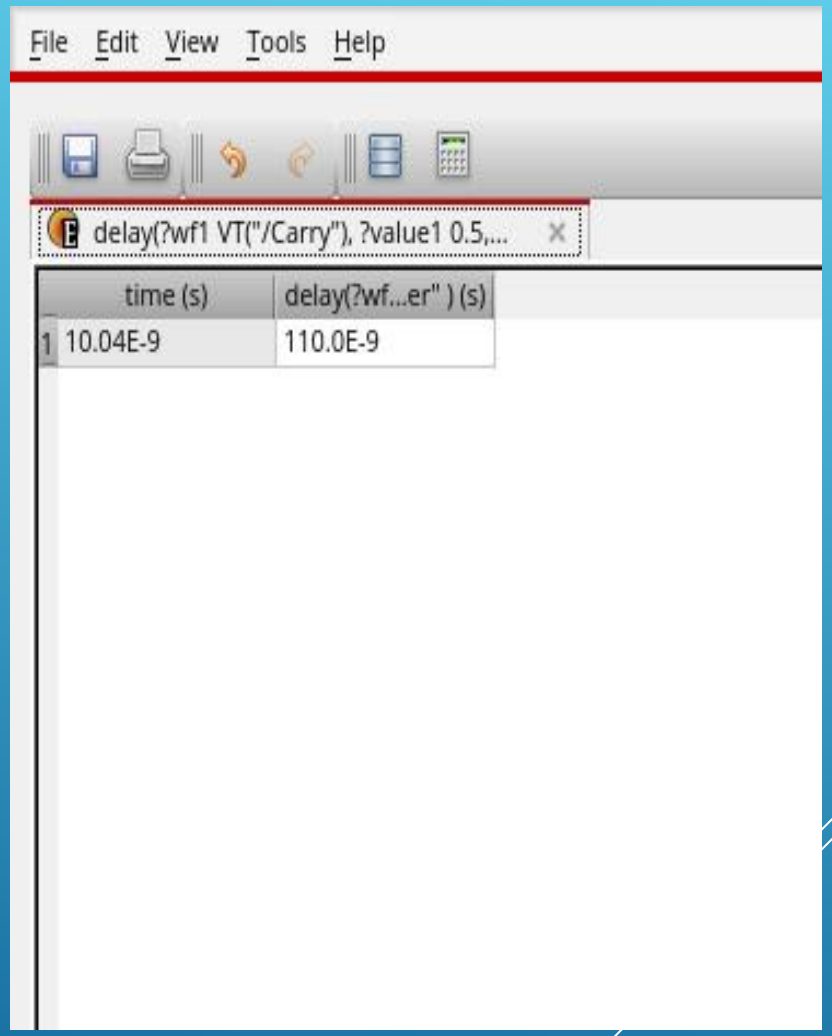


2)(II)POWER VALUE:



3)(II)DELAY VALUE:





CONCLUSION & INFERENCE

1)**Power Consumption:** Based on our observations after performing the experiments, we observe that the power consumed by the LCAFA3 and LCAFA4 circuits is exponentially lesser than the power consumed by the Static CMOS design of the full adder.

2) **Accuracy of Logic Levels:** When we look at the graphs of the circuits, we notice that the Static CMOS design is the most accurate by far, compared to the LCAFA3 & LCAFA4 designs. As a result, we only use the approximate FA designs only for error-tolerant applications, such as image processing.

3) **Applications:** Static CMOS designs are used where accuracy is necessary, such as our circuits in microprocessors or microcontrollers. LCAFA3 & LCAFA4 are only used where we can afford to tolerate errors to minimise the cost, such as image processing.